

Shijie Wang

shijie.wang@connect.polyu.hk | <https://sjay-wang.github.io>

Google Scholar: 2,100+ citations | h-index: 11

11 Yuk Choi Road, Hung Hom, Kowloon, Hong Kong

EDUCATION

PhD Candidate, The Hong Kong Polytechnic University

Advisor: Dr. Wenqi FAN (supervisor) and Prof. Qing LI (co-supervisor)

Hong Kong

08/2022–08/2026

BSc Information and Computing Science, University of Liverpool

Grade: First Class Honors Degree

Liverpool, UK

09/2018–07/2022

VISITING EXPERIENCE

Research Scholar, Institute of Data Science (IDS), National University of Singapore (NUS)

Advisor: Prof. See-Kiong Ng

Singapore

01/2026–08/2026

RESEARCH INTERESTS

- Large Language Models, Recommender Systems (RecSys), Retrieval-Augmented Generation (RAG), Graph Neural Networks (GNNs), Trustworthy AI.

PUBLICATION RECORDS

* indicates equal contribution.

Selected Publications:

- Shijie Wang**, Chengyi Liu, Yajuan Ding, Shanru Lin, See-Kiong Ng, Xu Xin, Wenqi Fan. “Mixture-of-Experts Knowledge Graph Retrieval-Augmented Generation for Multi-Agent LLM-based Recommendation”, In ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD**), 2026.
- Shijie Wang**, Wenqi Fan, Yue Feng, Xinyu Ma, Shuaiqiang Wang, Dawei Yin. “Knowledge Graph Retrieval-Augmented Generation for LLM-based Recommendation”, **ACL Main**, 2025.
- Shijie Wang**, Wenqi Fan, Xiao-yong Wei, Xiaowei Mei, Shanru Lin, Qing Li. “Multi-agent Attacks for Black-box Social Recommendations”, ACM Transactions on Information Systems (**TOIS**), 2024.
- Shijie Wang**, Jiani Huang, Wenqi Fan, Zhikai Chen, Yu Song, Wenzhuo Tang, Haitao Mao, Hui Liu, Xiaorui Liu, Dawei Yin, Qing Li. “Graph Machine Learning in the Era of Large Language Models (LLMs)”, ACM Transactions on Intelligent Systems and Technology (**TIST**), 2025.
- Jiani Huang, **Shijie Wang**, Liang-bo Ning, Wenqi Fan, Qing Li. “ReRec: Reasoning-Augmented LLM-based Recommendation Assistant via Reinforcement Fine-tuning”, **ACL Main**, 2026.
- Jiani Huang, **Shijie Wang**, Liang-bo Ning, Wenqi Fan, Shuaiqiang Wang, Dawei Yin, Qing Li. “Towards Next-Generation Recommender Systems: A Benchmark for Personalized Recommendation Assistant with LLMs”, **WSDM**, 2026.
- Zhuohang Jiang, Yuxin Chen, **Shijie Wang**, Haohao Qu, Zhou Jindong, Wenqi Fan, Li Qing, Dongxu Liang, Jun Wang. “Atomic Intent Reasoning: Bringing LLM Semantics to Industrial Cross-Domain Recommendations”, In ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD Industry**), 2026.
- Chengyi Liu, Xiao Chen, **Shijie Wang**, Wenqi Fan and Qing Li. “Continuous-time Discrete-space Diffusion Model for Recommendation”, **WSDM**, 2026.
- Lin Wang, **Shijie Wang**, Sirui Huang, Qing Li. “Simplifying Graph Neural Kernels: from Stacking Layers to Collapsed Structure”, **ICWL**, 2026.
- Liang-bo Ning*, **Shijie Wang***, Wenqi Fan, Qing Li, Xu Xin, Hao Chen, Feiran Huang. “CheatAgent: Attacking LLM-Empowered Recommender Systems via LLM Agent”, In ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD**), 2024.
- Chengyi Liu, Jiahao Zhang, **Shijie Wang**, Wenqi Fan, Qing Li. “Score-based Generative Diffusion Models for Social Recommendations”, In IEEE Transactions on Knowledge and Data Engineering (**TKDE**), 2025.
- Yuqi Zhang, Bin Guo, Nuo Li, Ying Zhang, **Shijie Wang**, Zhiwen Yu, Qing Li. “Tree-of-AdEditor: Heuristic Tree Reasoning for Automated Video Advertisement Editing with Large Language Model”, **IJCAI**, 2025.
- Jiahao Zhang, Lin Wang, **Shijie Wang**, Wenqi Fan. “Graph Unlearning with Efficient Partial Retraining”, In the Web Conference (**WWW PhD Symposium**), 2024.
- Yajuan Ding, Wenqi Fan, Liangbo Ning, **Shijie Wang**, Hengyun Li, Dawei Yin, Tat-Seng Chua, and Qing Li. “A Survey on RAG Meets LLMs: Towards Retrieval-Augmented Large Language Models”, In ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD**), 2024.
- Shijie Wang**, Shangbo Wang. “A Novel Multi-Agent Deep RL Approach for Traffic Signal Control”, In IEEE International Conference on Pervasive Computing and Communications Workshops (**PerCOM Workshops**), 2023.

Recent Preprints and Submissions:

- Wenqi Fan, Yi Zhou, **Shijie Wang**, Yuyao Yan, Hui Liu, Qian Zhao, Le Song, Qing Li. “Computational Protein Science in the Era of Large Language Models (LLMs)”, Under Review, 2025.

Tutorials:

- Wenqi Fan, Pangjing Wu, Yajuan Ding, Liangbo Ning, **Shijie Wang**, and Qing Li. “Towards Retrieval-Augmented Large Language Models: Data Management and System Design”, In IEEE International Conference on Data Engineering (**ICDE**, Tutorial), 2025.

- [2] Yujuan Ding, Wenqi Fan, Liangbo Ning, **Shijie Wang**, Hengyun Li, Dawei Yin, Tat-Seng Chua, and Qing Li. “RAG Meets LLMs: Towards Retrieval-Augmented Large Language Models”, In ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD**, Tutorial), 2024.
- [3] Yujuan Ding, **Shijie Wang**, Liangbo Ning, Qiaoyu Tan, Wenqi Fan, and Qing Li. “Recommender Systems in the Era of Large Language Models (LLMs)”, In International Joint Conference on Artificial Intelligence (**IJCAI**, Tutorial), 2024.
- [4] Wenqi Fan, Xiangyu Zhao, Lin Wang, Xiao Chen, Jingtong Gao, Qidong Liu, and **Shijie Wang**. “Trustworthy Recommender Systems: Foundations and Frontiers”, In ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD**, Tutorial), 2023.
- [5] Wenqi Fan, Xiangyu Zhao, **Shijie Wang**, Xiao Chen, Jingtong Gao, Qidong Liu, and Lin Wang. “Trustworthy Recommender Systems: Foundations and Frontiers”, In International Joint Conference on Artificial Intelligence (**IJCAI**, Tutorial), 2023.
- [6] Wenqi Fan, Xiangyu Zhao, Lin Wang, Xiao Chen, Jingtong Gao, Qidong Liu, and **Shijie Wang**. “Trustworthy Recommender Systems”, In The Web Conference (**WWW**, Tutorial), 2023.

PROFESSIONAL SERVICES

Conference Reviewer:

KDD’26, 25 (Outstanding Reviewer (top 10%)), 24, 23, AAAI’26, 25, 24, 23, ICDM’23, WWW’23, IJCAI’23, MM’23, WSDM’23, CIKM’22, LOG’22

Journal Reviewer:

ACM Transactions on Information Systems (TOIS)
 IEEE Transactions on Knowledge and Data Engineering (TKDE)
 IEEE Transactions on Information Forensics and Security (TIFS)
 IEEE Internet of Things Journal (IoT-J)
 ACM Transactions on Knowledge Discovery from Data (TKDD)

WORK EXPERIENCE

Baidu

Research Intern

06/2024–01/2025
 Search Science Team

Mentor: Dr. Dawei Yin and Dr. Shuaiqiang Wang

The Hong Kong Polytechnic University

Research Assistant

09/2022–08/2023
 Computing Department

TEACHING EXPERIENCE

The Hong Kong Polytechnic University

- | | |
|---|-------------|
| ◇ Teaching Assistant, COMP5511: Artificial Intelligence Concepts | Fall’2025 |
| ◇ Teaching Assistant, COMP3511: Legal Aspects and Ethics of Computing | Spring’2025 |
| ◇ Teaching Assistant, COMP5511: Artificial Intelligence Concepts | Fall’2024 |
| ◇ Teaching Assistant, COMP3511: Legal Aspects and Ethics of Computing | Spring’2024 |
| ◇ Teaching Assistant, COMP5511: Artificial Intelligence Concepts | Fall’2023 |